DYNAMICAL SYSTEMS
PROBLEM SET 4

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We begin by noting that all of the files referenced in this document are available at:
https://github.com/chtompki/MathFall2026/tree/main/DynamicalSystems/ProblemSet4

8.1.6 Consider the system $\dot{x} = y - 2x$, $\dot{y} = \mu + x^2 - y$.

a) Sketch the nullclines.
b) Find and classify the bifurcations that occur as μ varies.
c) Sketch the phase portrait as a function of μ.

Solution. **(a) Nullclines.** Setting each component of the vector field equal to zero gives

$$\dot{x} = 0: \quad y = 2x, \qquad \dot{y} = 0: \quad y = x^2 + \mu.$$

Thus the nullclines are a fixed straight line and an upward-opening parabola that moves vertically with μ. The direction of the flow is determined by

	condition	direction
$\dot{x} > 0$	$y > 2x$	right
$\dot{x} < 0$	$y < 2x$	left
$\dot{y} > 0$	$y < x^2 + \mu$	up
$\dot{y} < 0$	$y > x^2 + \mu$	down

The nullclines intersect when

$$x^2 - 2x + \mu = (x-1)^2 + \mu - 1 = 0.$$

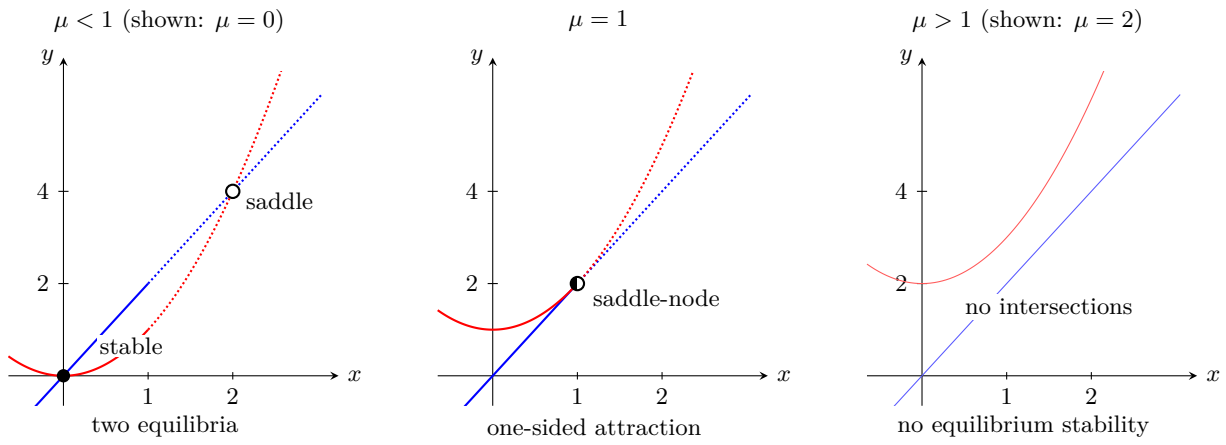
For $\mu < 1$, the two equilibria are

$$P_- = \big(1 - \sqrt{1-\mu},\, 2 - 2\sqrt{1-\mu}\big), \qquad P_+ = \big(1 + \sqrt{1-\mu},\, 2 + 2\sqrt{1-\mu}\big).$$

At $\mu = 1$ they coincide at $(1, 2)$, where the nullclines are tangent; for $\mu > 1$ there are no equilibria.

Nullcline sketches and equilibrium stability. The following sketches use $\mu = 0, 1, 2$ as representatives of the three cases. Blue denotes $y = 2x$ and red denotes $y = x^2 + \mu$. For $\mu \leq 1$, solid portions on the $x < 1$ side indicate the attracting equilibrium branch, and dotted portions on the $x > 1$ side indicate the saddle branch or the repelling side of the saddle-node. These line styles are visual guides to the classification at the intersections: points elsewhere on a nullcline are not equilibria and do not themselves have a stability classification. In particular, the dotted branch at the saddle does not mean that both eigen-directions are unstable.

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At $\mu = 1$, "one-sided attraction" refers to the center-manifold dynamics: the equilibrium attracts from the left and repels to the right, while the transverse direction is stable. It is therefore not a stable equilibrium in the Lyapunov sense. For $\mu > 1$, the curves are shown thin because there is no equilibrium to classify; calling either nullcline stable or unstable would be misleading.

(b) Jacobian and stability. The Jacobian is

$$J(x, y) = \begin{pmatrix} -2 & 1 \\ 2x & -1 \end{pmatrix}.$$

At the two equilibria, substitute $x = 1 \pm \sqrt{1-\mu}$ directly into the Jacobian:

$$J(P_\pm) = \begin{pmatrix} -2 & 1 \\ 2 \pm 2\sqrt{1-\mu} & -1 \end{pmatrix}.$$

Their traces, determinants, and characteristic polynomials are

$$\tau_\pm = -3, \qquad \Delta_\pm = -\big(\pm 2\sqrt{1-\mu}\big),$$

$$\det(\lambda I - J(P_\pm)) = \lambda^2 + 3\lambda - \big(\pm 2\sqrt{1-\mu}\big).$$

Here the upper sign corresponds to P_+ and the lower sign to P_-. We examine the two branches separately below. At the lower equilibrium P_-,

$$J_- = \begin{pmatrix} -2 & 1 \\ 2 - 2\sqrt{1-\mu} & -1 \end{pmatrix}, \qquad \Delta_- = 2\sqrt{1-\mu} > 0.$$

$$\lambda_{1,2}^{(-)} = \frac{-3 \pm \sqrt{9 - 8\sqrt{1-\mu}}}{2}.$$

Since the trace is negative and the determinant is positive, both eigenvalues have negative real parts. Hence P_- is asymptotically stable for every $\mu < 1$. More precisely,

parameter range	eigenvalues at P_-	classification
$\mu < -17/64$	complex, real part $-3/2$	stable spiral
$\mu = -17/64$	$\lambda_1 = \lambda_2 = -3/2$	improper stable node
$-17/64 < \mu < 1$	distinct, real, negative	stable node

Indeed, $9 - 8\sqrt{1-\mu} = 0$ gives $\sqrt{1-\mu} = 9/8$ and $\mu = 1 - (9/8)^2 = -17/64$. At this value $J_- \neq -\frac{3}{2}I$, so the repeated eigenvalue has only one independent eigenvector. This spiral-to-node transition does not change stability and is not a bifurcation of the local topological phase portrait.

At the upper equilibrium P_+,

$$J_+ = \begin{pmatrix} -2 & 1 \\ 2 + 2\sqrt{1-\mu} & -1 \end{pmatrix}, \qquad \Delta_+ = -2\sqrt{1-\mu} < 0.$$

$$\lambda_{1,2}^{(+)} = \frac{-3 \pm \sqrt{9 + 8\sqrt{1-\mu}}}{2}.$$

One eigenvalue is positive and one is negative, so P_+ is always an unstable saddle.

At $\mu = 1$,

$$J(1,2) = \begin{pmatrix} -2 & 1 \\ 2 & -1 \end{pmatrix}, \qquad \lambda_1 = 0, \quad \lambda_2 = -3.$$

Linearization alone cannot classify the zero-eigenvalue direction. Set

$$u = x - 1, \qquad w = y - 2x, \qquad \delta = \mu - 1.$$

Then the system becomes exactly

$$\dot{u} = w, \qquad \dot{w} = \delta + u^2 - 3w.$$

At $\delta = 0$, a center manifold $w = h(u)$ satisfies

$$h'(u)h(u) = u^2 - 3h(u).$$

Writing $h(u) = au^2 + O(u^3)$ and comparing quadratic terms gives $a = 1/3$. Thus the reduced equation is

$$\dot{u} = \tfrac{1}{3}u^2 + O(u^3).$$

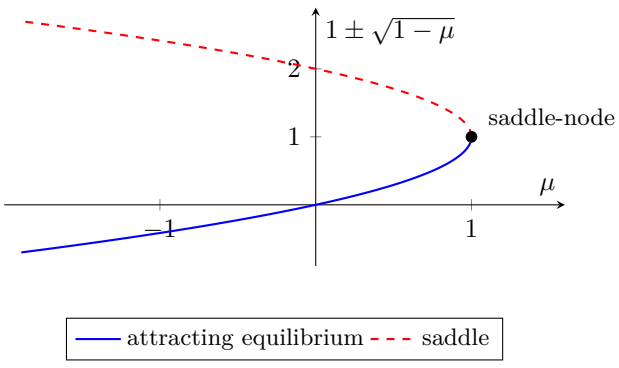
The equilibrium attracts along the center manifold from $u < 0$ and repels toward $u > 0$, while the transverse direction is attracting. In particular, it is not Lyapunov stable. Including the parameter gives the local saddle-node form

$$\dot{u} = \tfrac{1}{3}(\delta + u^2) + O\big(|u|^3 + |\delta u| + \delta^2\big).$$

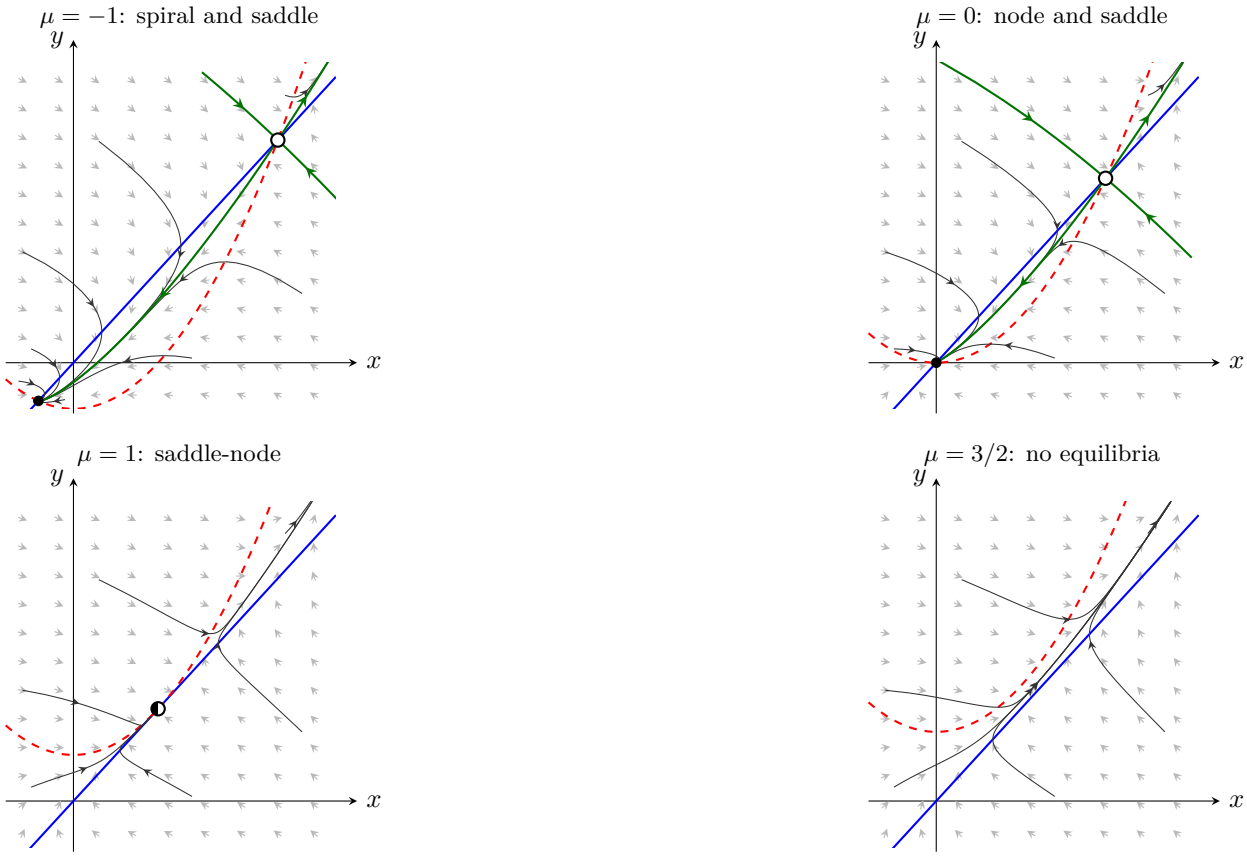
Therefore a stable equilibrium and a saddle collide and disappear at

$$\mu_c = 1 \quad \text{(saddle-node bifurcation)}.$$

No Hopf bifurcation is possible because the trace is always -3. Also, $\partial\dot{x}/\partial x + \partial\dot{y}/\partial y = -3$ everywhere, so Bendixson's criterion excludes periodic orbits in the plane.



(c) Phase portraits. The plots below show representative cases on either side of $\mu = 1$ and at the bifurcation. Blue solid curves are $\dot{x} = 0$, red dashed curves are $\dot{y} = 0$, and gray arrows give the flow direction. Black curves are numerically integrated trajectories; their arrows indicate increasing time. At the saddle, the green curves show its stable and unstable separatrices. A filled point marks the attractor, an open point the saddle, and a half-filled point the saddle-node.



For $\mu < 1$, nearby trajectories approach P_- (spiraling when $\mu < -17/64$), whereas the saddle's stable manifold separates trajectories heading toward the attractor from those escaping to the right. At $\mu = 1$, the two equilibria merge, leaving one-sided attraction and repulsion along the center direction. For $\mu > 1$ the parabola lies strictly above the line, and the flow passes through the region where the equilibria used to be. In the coordinates above,

$$\dot{w} = \mu - 1 + u^2 - 3w \geq \mu - 1 - 3w.$$

Thus, for any forward-complete trajectory, w is eventually positive and bounded away from zero, so $\dot{u} = w > 0$ eventually and u grows without bound. (Some trajectories may instead escape in finite time.) There is no remaining bounded attractor.

□

8.1.15 (The power of true believers) Sometimes a small band of unwavering advocates can win an entire population over to their point of view, as in the case of the civil rights or women's suffrage movements in the United States. Consider the following stylized model of such situations, which was studied by Marvel et al. (2012) and inspired by the earlier work of Xie et al. (2011).

The population is divided into four non-overlapping groups. An initially small group of true believers holds opinion A (for example, that women deserve the right to vote) and they are committed to this belief. Nothing that anyone says or does can change their minds. Another group of people currently agrees with them, but they are uncommitted to A. If someone argues for the opposing position, B, an uncommitted A-believer instantly becomes an AB, meaning someone who sees the merit in both positions. Likewise, B-believers instantly turn into members of the AB subpopulation when confronted with an argument for A. The people in the AB group, being fence-sitters, don't try to persuade anyone of anything. And they can be pushed to either side with the slightest nudge; when confronted with an argument for A—or for B—they join that camp. At each time step, we select two people at random and have one of them act as an advocate and the other as a listener. Assuming the members of the four groups mix with each other at random, the governing equations for the dynamics are

$$\dot{n}_A = (p + n_A)n_{AB} - n_A n_B,$$
$$\dot{n}_B = n_B n_{AB} - (p + n_A)n_{AB}.$$

where $n_{AB} = 1 - (p + n_A) - n_B$. Here the parameter p denotes the unchanging fraction of true believers in the population. The time-dependent variables n_A, n_B, and n_{AB} are the current fractions in the A, B, and AB subpopulations.

a) Interpret and justify the form of the various terms in the governing equations.
b) Assume that initially everyone believes in B, except for the true believers in A. Thus, $n_B(0) = 1 - p$, and $n_A(0) = n_{AB}(0) = 0$. Numerically integrate the system until it reaches equilibrium. Show that the final state changes discontinuously as a function of p. Specifically, show there is a critical fraction of true believers (call it p_c) such that for $p < p_c$, most people still accept B, whereas for $p > p_c$, everyone comes around to A.
c) Show analytically that $p_c = 1 - \sqrt{3}/2 \approx 0.134$. Thus, in this model, only about 13% of the population needs to be unwavering advocates to get everyone else to agree with them eventually.
d) What type of bifurcation occurs at p_c?

Solution. □

8.2.2 By calculating the linearization at the origin, show that the system $\dot{x} = -y + \mu x + xy^2$, $\dot{y} = x + \mu y - x^2$ has pure imaginary eigenvalues when $\mu = 0$.

Solution. □

8.2.3 (Computer work) By plotting phase portraits on the computer, show that the system $\dot{x} = -y + \mu x + xy^2$, $\dot{y} = x + \mu y - x^2$ undergoes a Hopf bifurcation at $\mu = 0$. Is it subcritical, supercritical, or degenerate?

Solution. □

8.2.4 (A heuristic analysis) The system $\dot{x} = -y + \mu x + xy^2$, $\dot{y} = x + \mu y - x^2$ can be analyzed in a rough, intuitive way as follows.

a) Rewrite the system in polar coordinates.
b) Show that if $r \ll 1$, then $\dot{\theta} \approx 1$ and $\dot{r} \approx \mu r + \frac{1}{8}r^3 + \cdots$, where the terms omitted are oscillatory and have essentially zero time-average around one cycle.
c) The formulas in part (b) suggest the presence of an unstable limit cycle of radius $r \approx \sqrt{-8\mu}$ for $\mu < 0$. Confirm that prediction numerically. (Since we assumed that $r \ll 1$, the prediction is expected to hold only if $|\mu| \ll 1$.)

The reasoning above is shaky. See Drazin (1992, pp. 188–190) for a proper analysis via the Poincaré-Lindstedt method.

Solution. □

8.2.10 (Bacterial respiration) Fairén and Velarde (1979) considered a model for respiration in a bacterial culture. The equations are

$$\dot{x} = B - x - \frac{xy}{1+qx^2}, \qquad \dot{y} = A - \frac{xy}{1+qx^2}$$

where x and y are the levels of nutrient and oxygen, respectively, and $A, B, q > 0$ are parameters. Investigate the dynamics of this model. As a start, find all the fixed points and classify them. Then consider the nullclines and try to construct a trapping region. Can you find conditions on A, B, q under which the system has a stable limit cycle? Use numerical integration, the Poincaré-Bendixson theorem, results about Hopf bifurcations, or whatever else seems useful. (This question is deliberately open-ended and could serve as a class project; see how far you can go.)

Solution. □

8.2.12 (Analytical criterion to decide if a Hopf bifurcation is subcritical or supercritical) Any system at a Hopf bifurcation can be put into the following form by suitable changes of variables:

$$\dot{x} = -\omega y + f(x,y), \qquad \dot{y} = \omega x + g(x,y),$$

where f and g contain only higher-order nonlinear terms that vanish at the origin. As shown by Guckenheimer and Holmes (1983, pp. 152–156), one can decide whether the bifurcation is subcritical or supercritical by calculating the sign of the following quantity:

$$\begin{aligned} 16a = {} & f_{xxx} + f_{xyy} + g_{xxy} + g_{yyy} \\ & + \frac{1}{\omega}\big[f_{xy}(f_{xx}+f_{yy}) - g_{xy}(g_{xx}+g_{yy}) - f_{xx}g_{xx} + f_{yy}g_{yy}\big] \end{aligned}$$

where the subscripts denote partial derivatives evaluated at $(0,0)$. The criterion is: If $a < 0$, the bifurcation is supercritical; if $a > 0$, the bifurcation is subcritical.

a) Calculate a for the system $\dot{x} = -y + xy^2$, $\dot{y} = x - x^2$.

b) Use part (a) to decide which type of Hopf bifurcation occurs for $\dot{x} = -y + \mu x + xy^2$, $\dot{y} = x + \mu y - x^2$ at $\mu = 0$. (Compare the results of Exercises 8.2.2–8.2.4.)

(You might be wondering what a measures. Roughly speaking, a is the coefficient of the cubic term in the equation $\dot{r} = ar^3$ governing the radial dynamics at the bifurcation. Here r is a slightly transformed version of the usual polar coordinate. For details, see Guckenheimer and Holmes (1983) or Grimshaw (1990).)

Solution. □

8.3.1 (Brusselator) The Brusselator is a simple model of a hypothetical chemical oscillator, named after the home of the scientists who proposed it. (This is a common joke played by the chemical oscillator community; there is also the "Oregonator," "Palo Altonator," etc.) In dimensionless form, its kinetics are

$$\begin{aligned} \dot{x} &= 1 - (b+1)x + ax^2y, \\ \dot{y} &= bx - ax^2y \end{aligned}$$

where $a, b > 0$ are parameters and $x, y \geq 0$ are dimensionless concentrations.

a) Find all the fixed points, and use the Jacobian to classify them.

b) Sketch the nullclines, and thereby construct a trapping region for the flow.
c) Show that a Hopf bifurcation occurs at some parameter value $b = b_c$, where b_c is to be determined.
d) Does the limit cycle exist for $b > b_c$ or $b < b_c$? Explain, using the Poincaré-Bendixson theorem.
e) Find the approximate period of the limit cycle for $b \approx b_c$.

Solution. □

8.4.4 (Second-order phase-locked loop) Using a computer, explore the phase portrait of

$$\ddot{\theta} + (1 - \mu \cos\theta)\dot{\theta} + \sin\theta = 0$$

for $\mu \geq 0$. For some values of μ, you should find that the system has a stable limit cycle. Classify the bifurcations that create and destroy the cycle as μ increases from 0.

Solution. □

8.5.4 (Logistic equation with sinusoidal harvesting) In Exercise 3.7.3 you were asked to consider a simple model of a fishery with constant harvesting. Now consider a generalization in which the harvesting varies periodically in time, perhaps due to daily or seasonal variations. To keep things simple, assume the periodic harvesting is purely sinusoidal (Benardete et al. 2008). Then, if the fish population grows logistically in the absence of harvesting, the model is given in dimensionless form by

$$\dot{x} = rx(1 - x) - h(1 + \alpha \sin t).$$

Assume that $r, h > 0$ and $0 < \alpha < 1$.

a) Show that if $h > r/4$ the system has no periodic solutions, even though the fish are being harvested periodically with period $T = 2\pi$. What happens to the fish population in this case?
b) By using a Poincaré map argument like that in the text, show that if $h < \dfrac{r}{4(1+\alpha)}$, there exists a 2π-periodic solution—in fact, a stable limit cycle—in the strip $1/2 < x < 1$. Similarly, show there exists an unstable limit cycle in the strip $0 < x < 1/2$. Interpret your results biologically.
c) What happens in between cases (a) and (b), i.e., for $\dfrac{r}{4(1+\alpha)} < h < \dfrac{r}{4}$?

Solution. □